A Giant Planet Vanished -- and It Isn't Coming Back for Decades

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For the first time, astronomers watched a giant planet vanish. The planet, a '**hot Jupiter**' called TOI-2535 b, sits about 1,000 **light-years** from Earth. Instead, it wobbled out of view.

Astronomers first spotted TOI-2535 b in April 2019 using NASA's **Transiting** Exoplanet Survey Satellite, known as TESS. The telescope finds planets by watching for tiny dips in a star's brightness when a planet passes in front of it. TOI-2535 b is a **gas giant** like Jupiter, but it hugs its star far more tightly than Mercury hugs our sun. Mercury takes 88 days to complete one orbit; TOI-2535 b needs just 8.25 days.

Over seven years of watching, researchers noticed something odd. The planet's crossings arrived as much as six hours earlier or later than predicted. Then, in March, the planet simply did not show up at all.

The researchers now believe TOI-2535 b has a close neighbor: another giant planet with a tilted orbit. The two planets tug on each other, wobbling like a pair of spinning tops until one tips out of line with Earth. Samuel Yee, a study co-author at the University of California, Los Angeles, called it the most extreme example researchers have seen. The planet's orbit is tilting so quickly, he said, that its **transit** disappeared while the team was watching.

The same kind of wobble shapes eclipses closer to home. The moon's orbit is tilted, so the moon, Earth, and sun line up only sometimes. Erik Petigura, a UCLA **astrophysicist** and study co-author, explained that if the moon's orbit lined up perfectly with Earth's, we would see an eclipse every month.

Based on their calculations, TOI-2535 b will not cross its star again for another 30 to 50 years. Its partner planet may appear sooner, once it emerges from behind its star.

Word Treasure

light-years -- A light-year is the distance light travels in one year, used to measure huge distances in space.

hot Jupiter -- A hot Jupiter is a giant planet like Jupiter that orbits very close to its star, so it is very hot.

gas giant -- A gas giant is a huge planet made mostly of gas, like Jupiter.

transit -- A transit is when a planet passes in front of its star, making the star look a little dimmer.

astrophysicist -- An astrophysicist is a scientist who studies stars, planets, and how the universe works.

Background you need

Astronomers find planets beyond our solar system by watching stars: when a planet passes in front of its star, the star's light dips slightly.

Planets can tug on each other with gravity, and over time those tugs can slowly tilt or shift their orbits.

The moon's orbit is tilted too, which is why the moon, Earth, and sun only line up for an eclipse some of the time.

Because TOI-2535 b is so far away, astronomers can only study it by watching its star, so a missing crossing tells them something has changed.

Break it down

LEAD: For the first time, astronomers watched a giant planet vanish as it wobbled out of view.

KEY EVENT: TOI-2535 b, a hot Jupiter 1,000 light-years away, failed to cross its star in March.

+- WHO: Astronomers using NASA's TESS satellite, including Samuel Yee and Erik Petigura of UCLA.

+- WHEN: First spotted in April 2019 and watched for seven years; the crossing vanished in March.

+- CLUE: Its crossings had arrived up to six hours early or late.

EXPLANATION: A neighboring giant planet with a tilted orbit tugs on it, tipping it out of line with Earth.

OPEN QUESTION: When will its partner planet appear, and what will further watching reveal about the pair?

Why it matters

Watching a planet disappear in real time shows how planets shape each other's paths, and it gives astronomers a rare live look at the forces that move worlds around their stars.

Test what you remember

1. What is TOI-2535 b?

- (A) A comet passing Earth
- (B) A hot Jupiter about 1,000 light-years from Earth
- (C) A star near Mercury
- (D) A moon of Jupiter

2. How was TOI-2535 b first spotted?

- (A) By a radio dish in 2010
- (B) By NASA's TESS satellite in April 2019
- (C) By a telescope on the moon
- (D) By astronauts on the space station

3. How long does TOI-2535 b take to orbit its star?

- (A) 365 days
- (B) 8.25 days
- (C) 88 days
- (D) 30 to 50 years

4. What happened in March?

- (A) The planet did not show up for its crossing at all
- (B) The planet split in two
- (C) The planet moved closer to Earth
- (D) The planet crashed into its star

5. What do researchers think is tilting TOI-2535 b's orbit?

- (A) A storm on the planet
- (B) Another giant planet with a tilted orbit
- (C) A black hole nearby
- (D) The moon's gravity

6. When will TOI-2535 b cross its star again?

- (A) In about 8 days
- (B) In 30 to 50 years
- (C) Next March
- (D) Never again

Answer key (try the quiz first!): 1: B 2: B 3: B 4: A 5: B 6: B

Questions to think about

1. **Samuel Yee: He calls it the most extreme example researchers have seen, with the orbit tilting so fast the transit vanished while the team watched.**

2. **Erik Petigura: He explains the same kind of tilt with the moon: if the moon's orbit lined up perfectly with Earth's, we would see an eclipse every month.**

MY THOUGHTS (at least 20 words):